

Data Universe and Sample Construction

90K

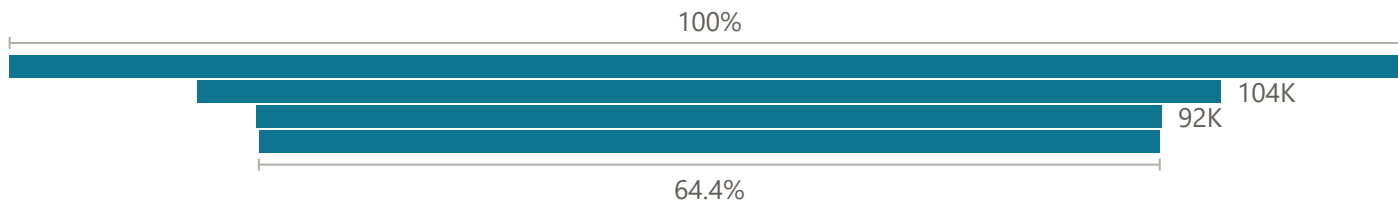
Company Count

13

Count of RatioLabel

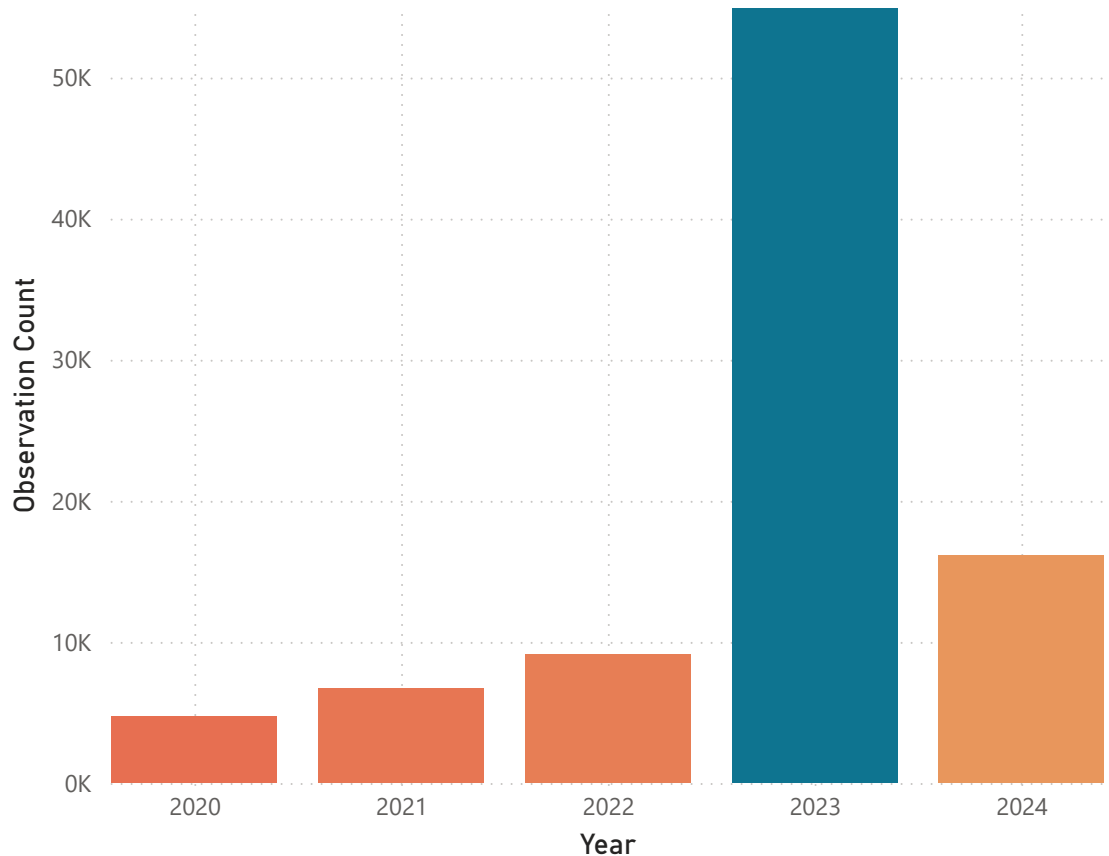
Sample Construction Funnel

1. Raw source records
2. German raw records
3. Clean active German base
4. Final ratio-ready sample



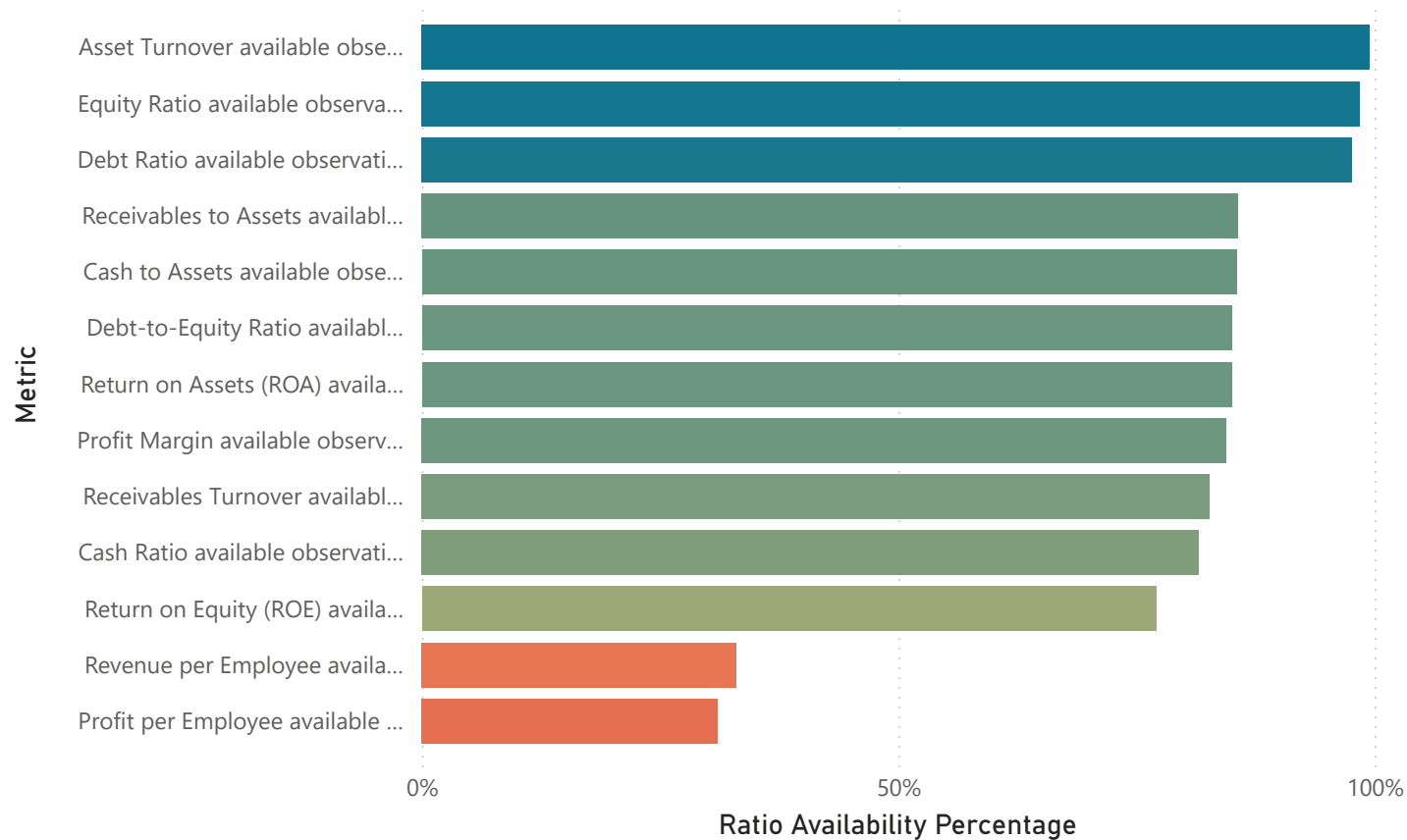
Final Analytical Observation by Year

Observation Count 4.74K 29.83K 54.91K



Ratio Availability Percentage by Metric

Ratio Availability Percentage 31.05% 65.27% 99.49%

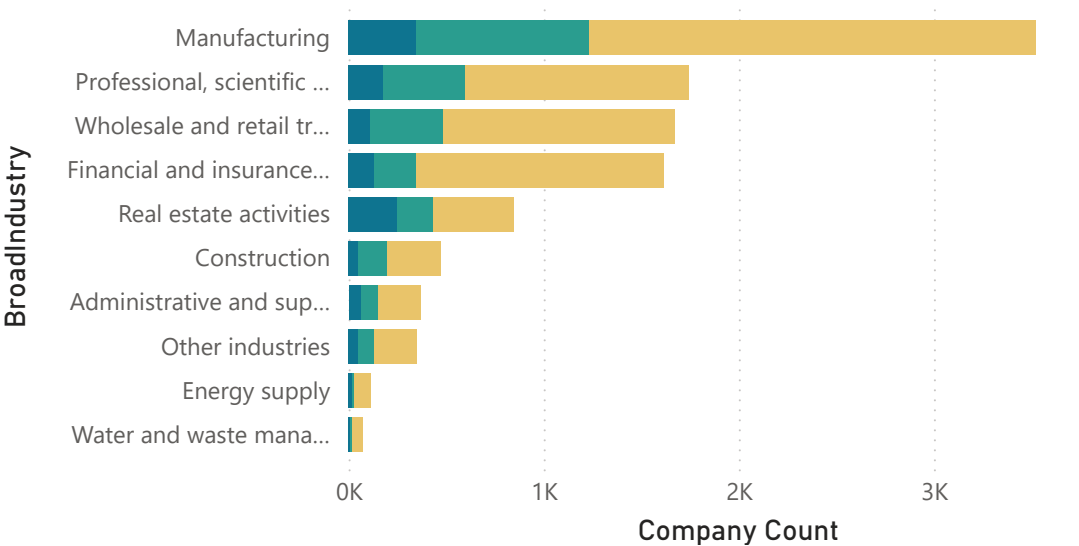


02 - Executive Financial Overview

- Broad_Industry
- ☐ Administrative and support services
 - ☐ Construction
 - ☐ Energy supply
 - ☐ Financial and insurance activities
 - ☐ Manufacturing
 - ☐ Other industries
 - ☐ Professional, scientific and technical activities
 - ☐ Real estate activities
 - ☐ Unknown
 - ☐ Water and waste management
 - ☐ Wholesale and retail trade

Profitability Composition by Industry

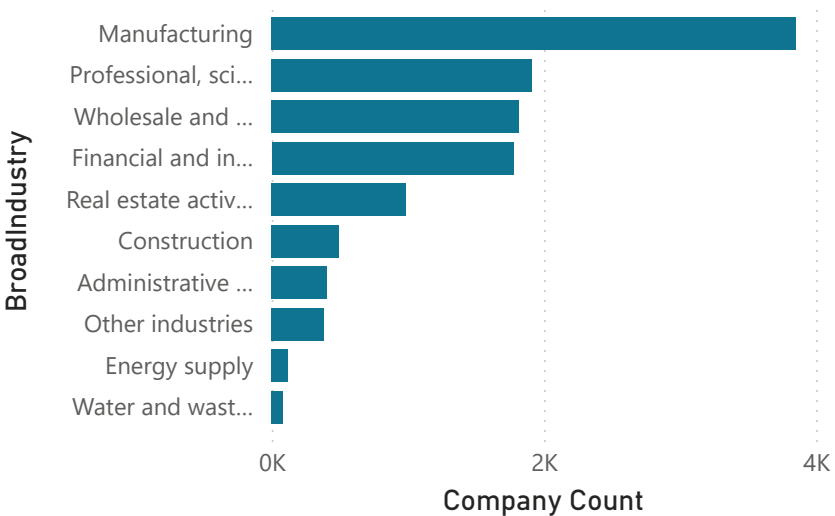
ProfitabilityStatus Break-even Loss-making Profitable



BroadIndustry	Ave. Profit Margin	Ave. ROA	Ave. ROE	Ave. equity_ratio
Wholesale and retail trade	0.02	0.05	0.15	0.3
Water and waste management	0.07	0.08	0.23	0.4
Unknown	0.20	0.15	0.28	0.4
Real estate activities	0.09	0.04	0.11	0.3
Professional, scientific and technical activities	0.04	0.06	0.14	0.4
Other industries	0.08	0.07	0.15	0.3

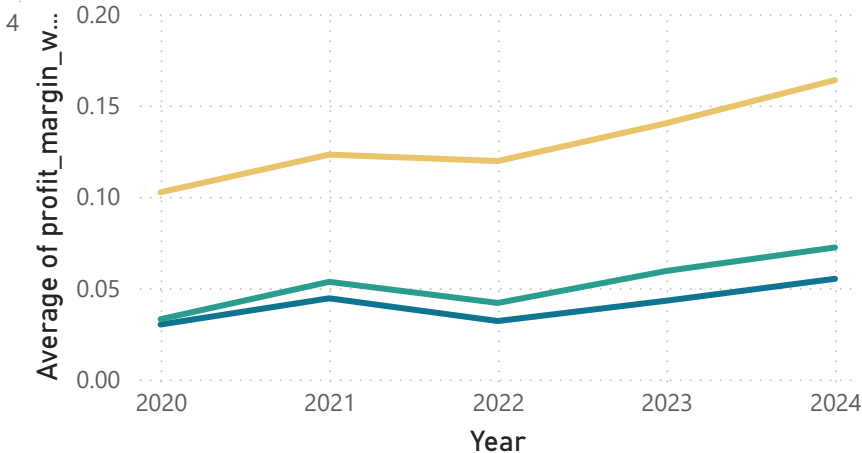
- Year
- ☐ 2020
 - ☐ 2021
 - ☐ 2022
 - ☐ 2023
 - ☐ 2024
- EmployeeSizeClass
- ☐ Large (250+)
 - ☐ Medium (50–249)
 - ☒ Micro (0–9)
 - ☐ Small (10–49)
 - ☐ Unknown

Company Count by BroadIndustry



Average Profit margin, ROA & ROE

Average of profit_margin_w... Average of roa_win Average of roe...



12K

Company Count

0.04

Ave. Profit margin

0.06

Ave. ROA

0.14

Ave. ROE

0.60

(%)Profitable Comp. Sh...

0.42

Ave. equity ratio

03 - Profitability Landscape

Broad_Industry

- ☐ Administrative and support services
- ☐ Construction
- ☐ Energy supply
- ☐ Financial and insurance activities
- ☐ Manufacturing
- ☐ Other industries
- ☐ Professional, scientific and technical services
- ☐ Real estate activities
- ☐ Unknown
- ☐ Water and waste management
- ☐ Wholesale and retail trade

Year

- ☐ 2020
- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024

EmployeeSizeClass

- ☐ Large (250+)
- ☐ Medium (50–249)
- ☐ Micro (0–9)
- ☐ Small (10–49)
- ☐ Unknown

Asset Turnover, Profit margin and Company Count by BroadIndustry



3.45

Average of asset_turno...

0.05

Ave. Profit margin

0.06

Ave. ROA

0.16

Ave. ROA

0.21

Loss-Making Share

0.57

(%)Profitable Comp. Sh...

04 - Industry Benchmarking

- RatioGroup
- ☐

Capital Structure
- ☐

Efficiency
- ☐

Liquidity
- ☐

Productivity
- ☐

Profitability
- ☐

Working Capital

- RatioLabel
- ☐

Asset Turnover
- ☐

Cash Ratio
- ☐

Cash to Assets
- ☐

Debt Ratio
- ☐

Debt-to-Equity Ra...
- ☐

Equity Ratio
- ☐

Profit Margin
- ☐

Profit per Employee
- ☐

Receivables to As...
- ☐

Receivables Turno...
- ☐

Return on Assets (...)
- ☐

Return on Equity (...)
- ☐

Revenue per Empl...

Select one ratio

Selected Ratio Label

Construction

Highest Industry

Professional, scientif...

Lowest Industry

BroadIndustry	AvailableN	AvailabilityPct	PositiveSharePct
Real estate activities	6244	70.02	41.93
Real estate activities	6378	71.52	42.88
Real estate activities	5382	60.35	45.97
Real estate activities	1181	13.24	50.13
Construction	548	54.64	60.77
Administrative and support services	2620	85.40	62.48
Administrative and support services	2635	85.89	62.73
Administrative and support services	962	31.36	64.45
Administrative and support services	2377	77.48	65.92

9

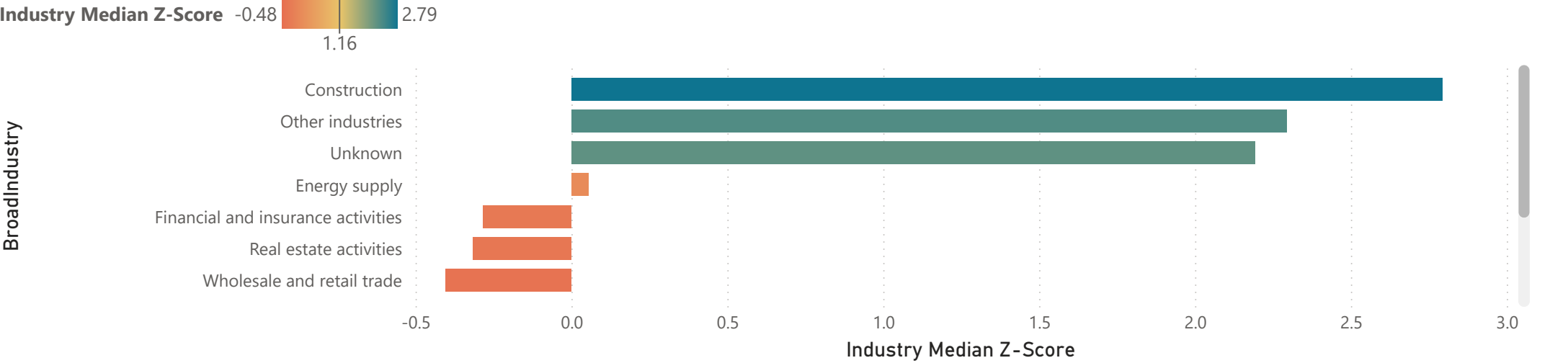
Main Industry Count

89K

Main Industry Companies

This page benchmarks industries based on the selected financial ratio. The median z-score shows whether an industry is above or below the overall sample median after standardisation. Positive values indicate that the industry performs above the average benchmark for the selected ratio, while negative values indicate below-average performance. This helps compare industries even when ratios have different scales.

Industry Median Z-Score, Max of AvailableN, Sum of AvailabilityPct, Max of IQR and Max of DistinctCompanies by BroadIndustry



05 - Financial Structure, Liquidity and Efficiency

Broad_Industry

- ☐ Administrative and support services
- ☐ Construction
- ☐ Energy supply
- ☐ Financial and insurance activities
- ☐ Manufacturing
- ☐ Other industries
- ☐ Professional, scientific and technical activities
- ☐ Real estate activities
- ☐ Unknown
- ☐ Water and waste management
- ☐ Wholesale and retail trade

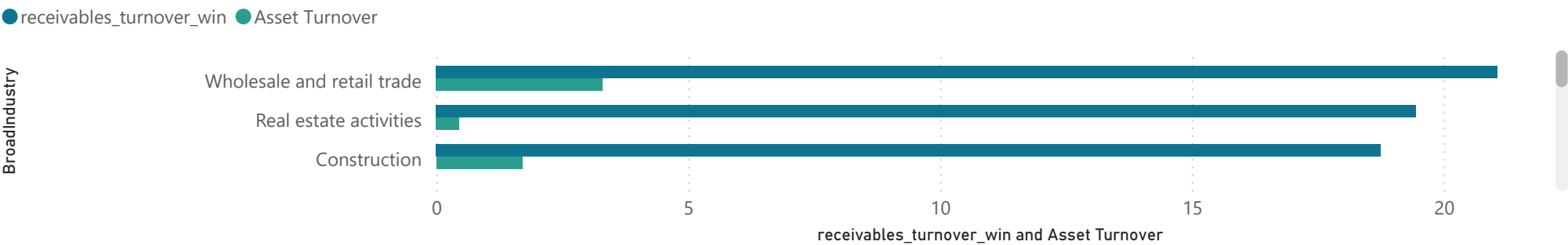
Year

- ☐ 2020
- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024

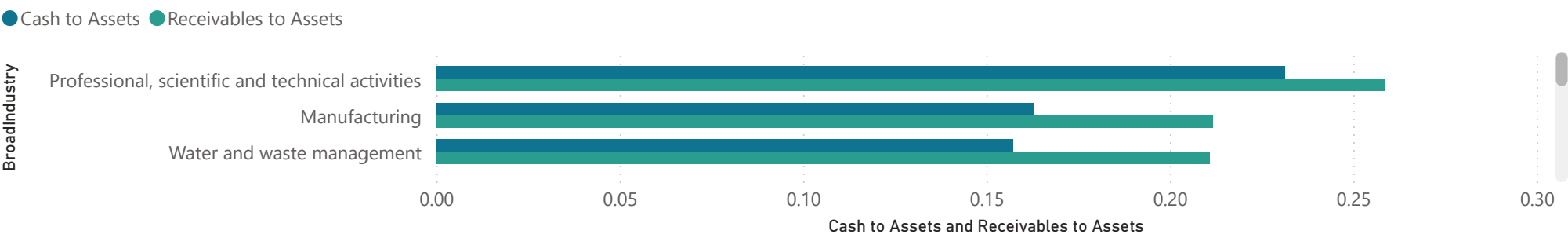
Employee Size Class

- ☐ Large (250+)
- ☐ Medium (50–249)
- ☐ Micro (0–9)
- ☐ Small (10–49)
- ☐ Unknown

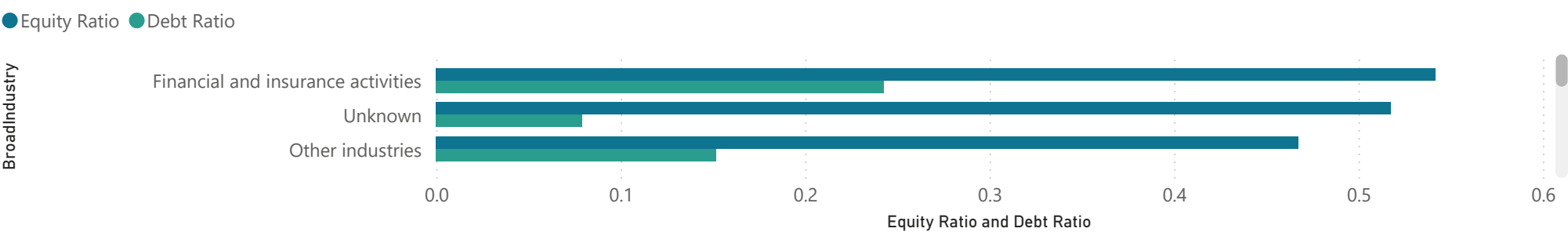
Receivables turnover and Asset Turnover by BroadIndustry



Liquidity and Working Capital Structure



Capital Structure by Industry



0.43

Median Equity Ratio

0.37

Median Debt Ratio

0.60

Median Debt-to-Equity

0.13

Median Cash to Assets

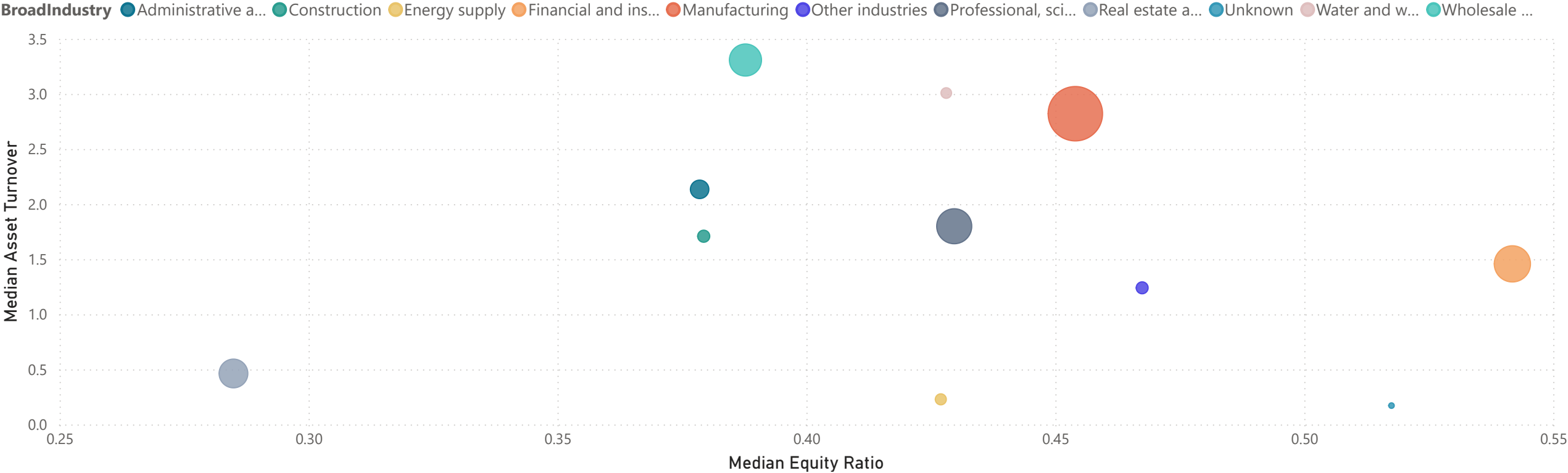
2.24

Median Asset Turnover

366.36K

Median of revenue_per...

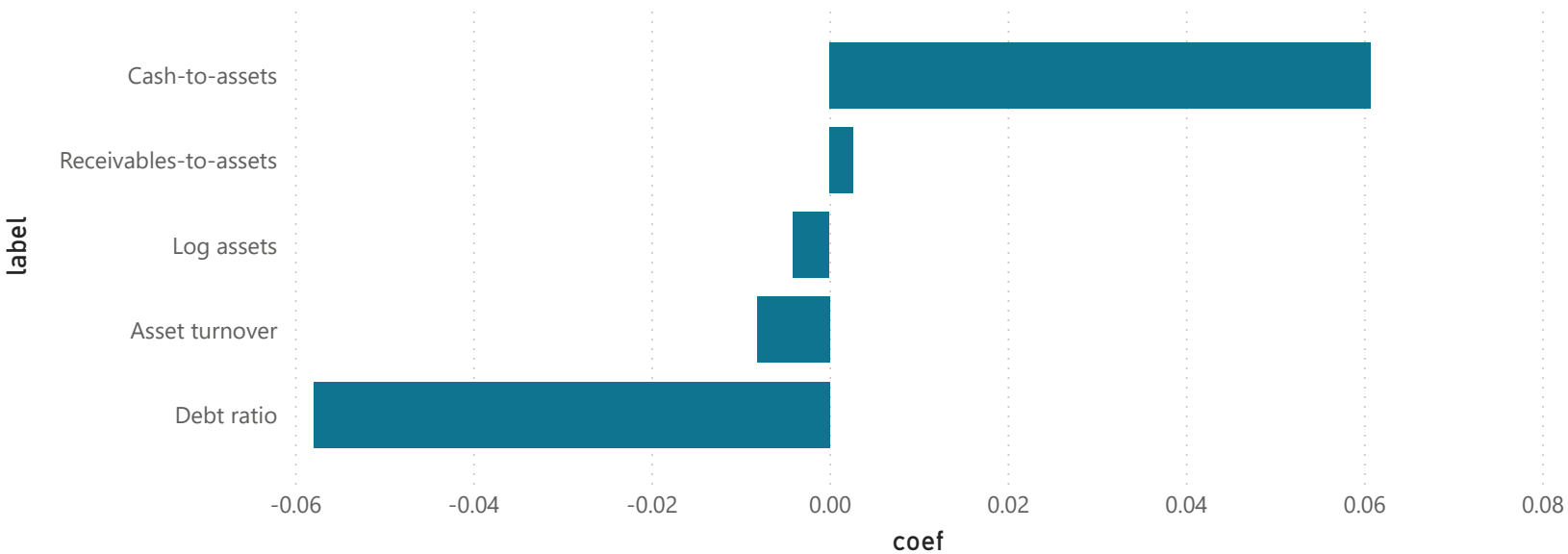
Median Equity Ratio, Median Asset Turnover and Company Count by BroadIndustry



BroadIndustry	Company Count	Median Revenue per Employee	Median Profit per Employee	Median Equity Ratio	Median Asset Turnover
Construction	984	14,000,000.00	22,410.28	0.38	1.71
Other industries	944	11,901,029.56	52,543.79	0.47	1.24
Unknown	47	11,475,382.17	408,029.40	0.52	0.17
Energy supply	744	2,500,000.00	95,787.15	0.43	0.23
Financial and insurance activities	14662	1,081,810.59	22,812.00	0.54	1.46
Real estate activities	8704	950,000.00	10.12	0.28	0.46
Wholesale and retail trade	11201	580,000.00	5,073.67	0.39	3.31
Administrative and support services	3033	406,250.00	4,158.41	0.38	2.13
Water and waste management	662	357,419.35	5,244.85	0.43	3.01
Manufacturing	35676	294,444.44	3,142.15	0.45	2.82
Professional, scientific and technical activities	13606	266,190.48	4,199.10	0.43	1.80

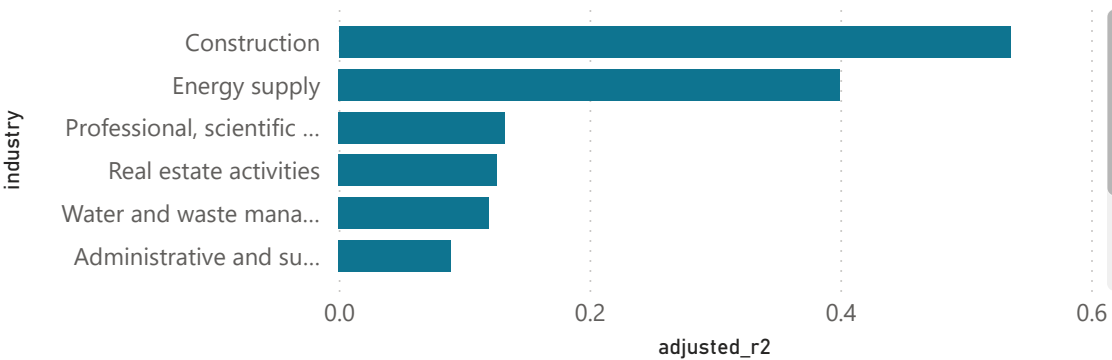
06 - Financial Structure, Liquidity and Efficiency

Which factors affect profit margin?



The coefficient chart indicates that cash-to-assets is the strongest positive profitability signal, while debt ratio is the strongest negative signal. Asset turnover is also negatively associated with profit margin, supporting the margin-turnover trade-off. Receivables-to-assets and firm size show weaker effects and should be interpreted cautiously.

adjusted_r2 by industry



Profitability Driver Strength by Industry

industry	Asset turnover	Cash to assets	Debt ratio	Receivables to assets
Administrative and support services	-0.11	0.12	-0.18	-0.03
Construction	-0.88	0.02	-0.19	0.15
Energy supply	-0.34	0.11	-0.03	0.02
Financial and insurance activities	-0.10	0.12	-0.14	-0.11
Manufacturing	-0.11	0.12	-0.16	0.05
Professional, scientific and technical activities	-0.18	0.13	-0.24	0.03
Real estate activities	-0.30	0.07	-0.22	0.09
Water and waste management	-0.41	0.12	-0.09	0.17
Wholesale and retail trade	-0.16	0.15	-0.10	0.07

This heatmap shows that profitability drivers are not equally strong across industries. Debt ratio is mostly negative, cash-to-assets is mostly positive, and asset turnover is often negative. This supports the idea that profitability analysis should be industry-specific rather than based on one universal rule.

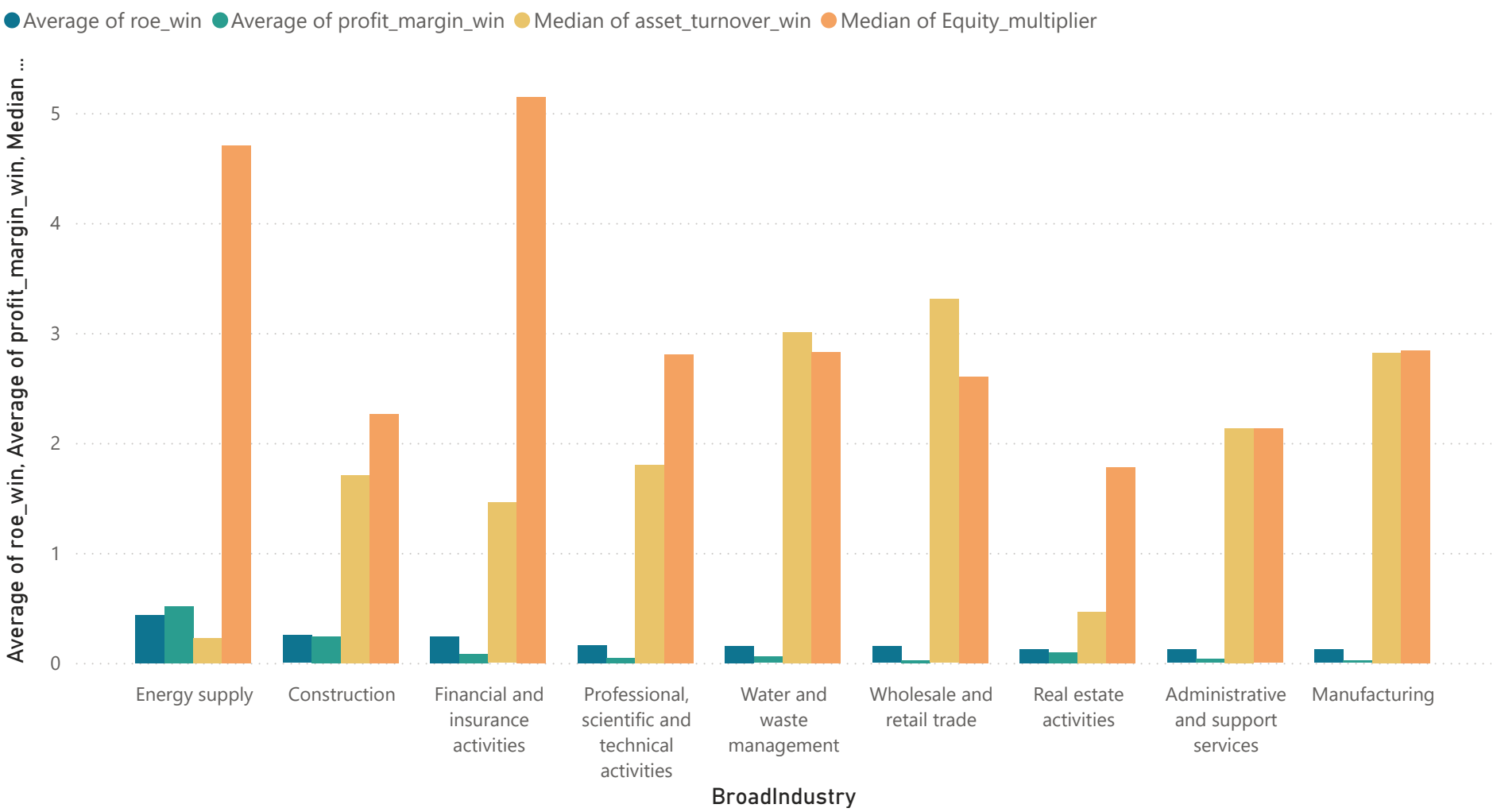
07 - Dupont Analysis

Broad_Industry

ROE Components: Margin, Efficiency and Leverage

- ☐ Administrative and support servi...
- ☐ Construction
- ☐ Energy supply
- ☐ Financial and insurance activities
- ☐ Manufacturing
- ☐ Other industries
- ☐ Professional, scientific and techn...
- ☐ Real estate activities
- ☐ Unknown
- ☐ Water and waste management
- ☐ Wholesale and retail trade

- Year
- ☐ 2020
 - ☐ 2021
 - ☐ 2022
 - ☐ 2023
 - ☐ 2024
- EmployeeSizeClass
- ☐ Large (250+)
 - ☐ Medium (50–249)
 - ☐ Micro (0–9)
 - ☐ Small (10–49)
 - ☐ Unknown



0.05

Profit Margin

2.24

Asset Turnover

0.43

Equity_ratio

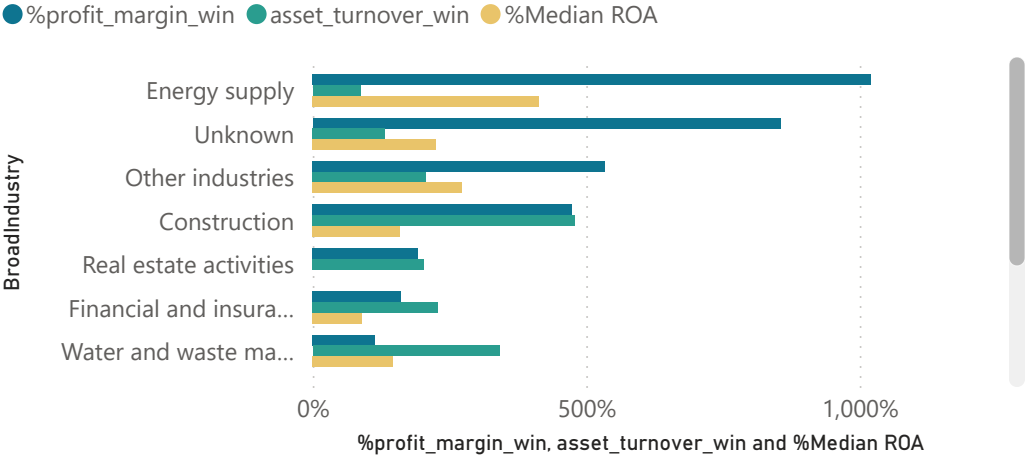
0.03

ROA

0.09

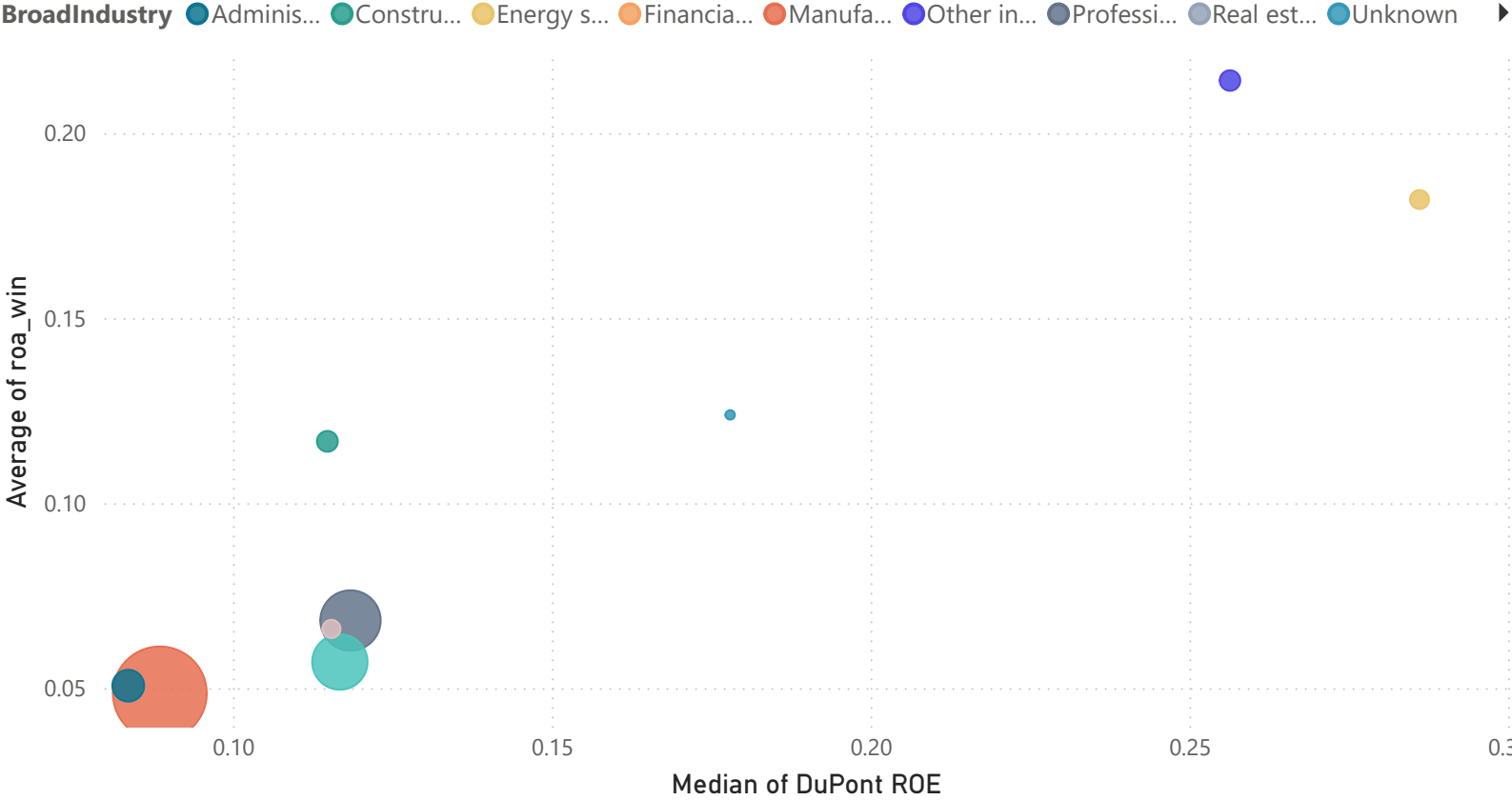
ROE

ROA Components: Margin and Asset Efficiency

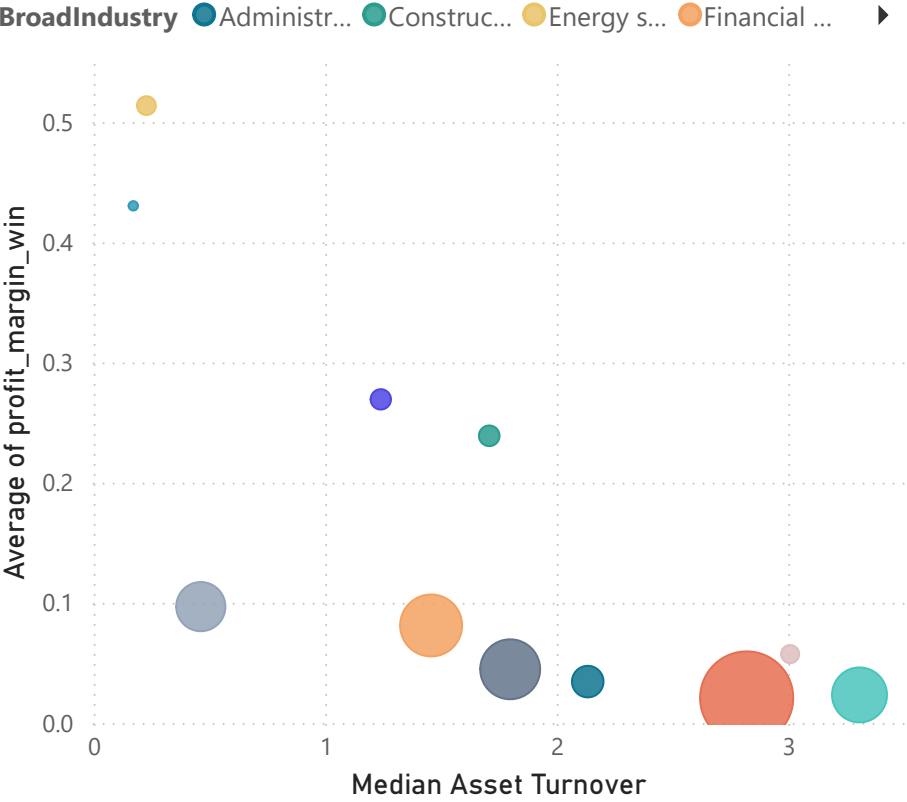


The DuPont analysis shows that industry profitability differences are not driven by one single profitability measure. ROA reflects both profit margin and asset turnover, while ROE additionally reflects leverage through the equity multiplier. This helps distinguish high-margin industries from high-efficiency industries and prevents misleading interpretation of ROE as a stand-alone performance indicator.

Median of DuPont ROE, Average of roa_win and Company Count by BroadIndustry and BroadIndustry



Median Asset Turnover, Average of profit_margin_win and Company Count by BroadIndustry



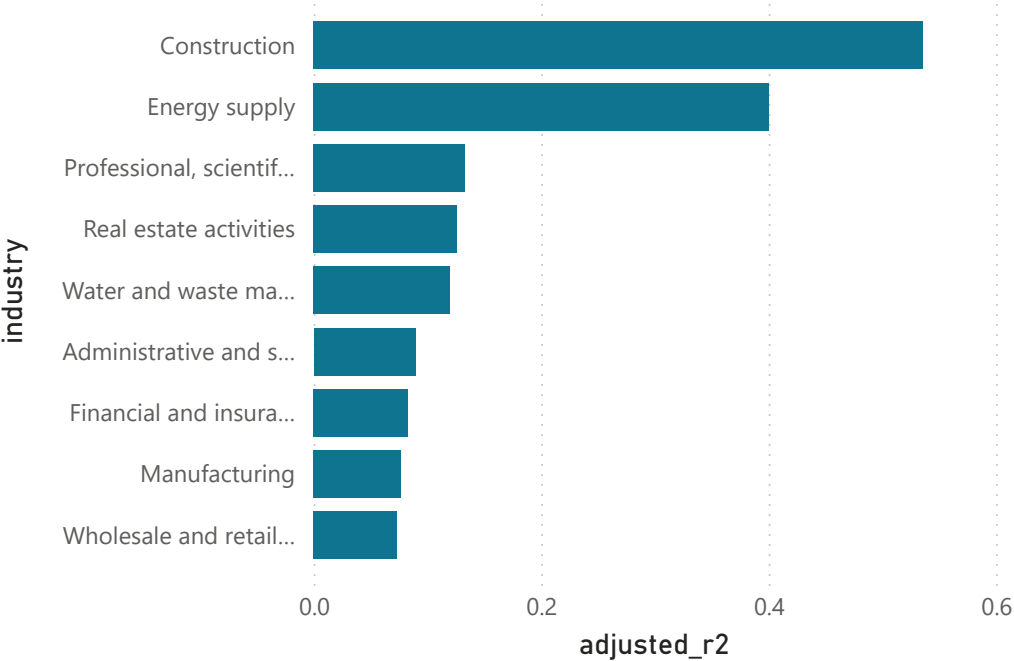
08 - Industry-Specific Profitability Drivers

How debt, liquidity, receivables and asset turnover differ across industries

Metric	Value
Industries modelled	9
Main Stable driver	Debt ratio negative
Strongest adjusted R2	0.536
Strongest model fit	Construction
Weakest adjusted R2	0.073
Weakest model fit	Wholesale and retail

Construction and energy have much higher industry-level explanatory power, while manufacturing and wholesale/retail are harder to explain using only the selected ratios.

how well the profitability Model Fits Each Industry



The selected financial ratios explain profitability much better in construction and energy than in manufacturing or wholesale/retail, suggesting that profitability mechanisms are industry-specific.

industry	Asset turnover	Cash to assets	Debt ratio	Receivables to assets
Administrative and support services	-0.11	0.12	-0.18	-0.03
Construction	-0.88	0.02	-0.19	0.15
Energy supply	-0.34	0.11	-0.03	0.02
Financial and insurance activities	-0.10	0.12	-0.14	-0.11
Manufacturing	-0.11	0.12	-0.16	0.05
Professional, scientific and technical activities	-0.18	0.13	-0.24	0.03
Real estate activities	-0.30	0.07	-0.22	0.09
Water and waste management	-0.41	0.12	-0.09	0.17
Total	-0.29	0.11	-0.15	0.05

Interpretation: The direction of profitability drivers is not identical across all industries. Debt ratio is mostly negatively associated with profit margin, while cash-to-assets is mostly positive. Asset turnover is often negative, supporting the margin–turnover trade-off. Receivables-to-assets is less stable and should be treated cautiously. The adjusted R² results show that the same financial ratios explain profitability better in some industries than others.

Profitability drivers are not universal. Industry context matters.

The industry-specific models show that profitability mechanisms differ across sectors. Although debt ratio, cash-to-assets and asset turnover are stable overall signals, their strength varies substantially by industry, confirming the need for industry-aware financial-ratio analysis.

industry	ci_lower	ci_upper	Observations	standardized_beta	predictor_label	p_value
Administrative and support services	-0.17	-0.05	2445	-0.11	Asset turnover	0.00
Administrative and support services	0.05	0.19	2445	0.12	Cash to assets	0.00
Administrative and support services	-0.22	-0.13	2445	-0.18	Debt ratio	0.00
Administrative and support services	-0.08	0.03	2445	-0.03	Receivables to assets	0.32
Construction	-0.94	-0.81	822	-0.88	Asset turnover	0.00
Construction	-0.04	0.07	822	0.02	Cash to assets	0.59
Construction	-0.24	-0.14	822	-0.19	Debt ratio	0.00
Construction	0.08	0.21	822	0.15	Receivables to assets	0.00
Energy supply	-0.49	-0.19	603	-0.34	Asset turnover	0.00
Energy supply	0.04	0.17	603	0.11	Cash to assets	0.00
Energy supply	-0.10	0.04	603	-0.03	Debt ratio	0.41
Energy supply	-0.05	0.08	603	0.02	Receivables to assets	0.66
Financial and insurance activities	-0.11	-0.08	9163	-0.10	Asset turnover	0.00
Financial and insurance activities	0.09	0.15	9163	0.12	Cash to assets	0.00
Financial and insurance activities	-0.16	-0.12	9163	-0.14	Debt ratio	0.00
Financial and insurance activities	-0.13	-0.09	9163	-0.11	Receivables to assets	0.00
Manufacturing	-0.12	-0.10	30423	-0.11	Asset turnover	0.00
Manufacturing	0.10	0.14	30423	0.12	Cash to assets	0.00
Manufacturing	-0.18	-0.15	30423	-0.16	Debt ratio	0.00
Manufacturing	0.04	0.07	30423	0.05	Receivables to assets	0.00
Professional, scientific and technical activities	-0.20	-0.15	10899	-0.18	Asset turnover	0.00
Professional, scientific and technical activities	0.10	0.16	10899	0.13	Cash to assets	0.00
Professional, scientific and technical activities	-0.26	-0.21	10899	-0.24	Debt ratio	0.00
Professional, scientific and technical activities	0.01	0.06	10899	0.03	Receivables to assets	0.01
Real estate activities	-0.36	-0.24	5314	-0.30	Asset turnover	0.00
Real estate activities	0.04	0.11	5314	0.07	Cash to assets	0.00
Real estate activities	-0.25	-0.19	5314	-0.22	Debt ratio	0.00
Real estate activities	0.04	0.14	5314	0.09	Receivables to assets	0.00
Water and waste management	-0.59	-0.23	589	-0.41	Asset turnover	0.00
Water and waste management	0.00	0.24	589	0.12	Cash to assets	0.06
Water and waste management	-0.19	0.01	589	-0.09	Debt ratio	0.09
Water and waste management	0.02	0.31	589	0.17	Receivables to assets	0.03
Wholesale and retail trade	0.18	0.14	6176	0.16	Asset turnover	0.00

Final Insights from the Financial Ratio Analysis

Profitability differs strongly across industries.:

Industry context matters because the same financial ratios do not explain profitability equally well in every sector.

Debt ratio is mostly a negative profitability signal.

Higher debt ratio is generally associated with lower profit margin, especially after controlling for industry and year effects.

Cash-to-assets is mostly a positive profitability signal.

Firms with stronger liquidity positions tend to show stronger profit margins

DuPont analysis shows that ROA and ROE should not be interpreted alone. Profitability can come from margin strength, asset efficiency or leverage. Therefore, ROE may look strong even when it is partly driven by higher leverage.

Overall conclusion:

The analysis shows that profitability is not driven by one universal ratio. Debt ratio, cash-to-assets and asset turnover are the most stable profitability signals, but their strength differs across industries. Therefore, financial-ratio analysis should be interpreted with industry context, DuPont decomposition and regression-based controls rather than simple standalone ratios.

Important caution: The results show adjusted associations, not causal effects. This means the analysis can identify which financial ratios are linked with profitability, but it cannot prove that these ratios directly cause higher or lower profitability.

The thesis finds that profitability patterns among German firms are industry-dependent. While liquidity and capital structure show stable relationships with profit margin, the explanatory power of financial ratios varies considerably across sectors. This supports the use of industry-aware and model-based financial-ratio analysis instead of relying only on simple descriptive comparisons.